

PLATFORM OVERVIEW

A production-ready e-commerce backend, storefront, and admin panel — designed as a foundation for custom-code projects, not as another SaaS rental.

STATUS

Production-ready core

First merchant launch · May 2026

DOCUMENT

v2 · May 2026

STATUS

Production-ready core. First merchant launch scheduled mid-to-late May 2026. End-to-end QA in progress on real fiscal, payment, and refund flows. The platform ships with infrastructure most ecommerce builds bolt on later — broadcast email engine, incident notification system, fuzzy CSV import, idempotent refund pipeline, audit trails, encrypted token storage.

WHAT IT IS

A complete e-commerce backend + storefront + admin panel a merchant can run on their own infrastructure. No per-transaction platform fee, no SaaS subscription, no app-store tax. The code base is owned by the merchant (or licensed); operating cost is hosting + payment processing fees + email-sending only.

Locale-agnostic core, EU-style fiscal compliance built in (configurable per shop). All user-facing content — pages, emails, error messages — lives as editable templates. Translation to any market is a content task, not an engineering one.

CUSTOMER-FACING STOREFRONT

Browse & discover

- Product catalog with category hierarchy (multi-level), producer/brand filtering, full-text search, on-sale highlights
- SEO-friendly URLs with category landing pages (configurable URL prefix per locale) and per-product slugs
- Per-category and per-product meta tags (title, description, keywords) — SEO data preserved on import from existing platforms
- AI SEO evaluation for product titles, descriptions, metadata, LLM readability, and AI crawler discoverability before publish
- Mobile-first responsive storefront
- Recently-viewed products tracking
- Wishlist (persisted across sessions for logged-in users)

Cart & checkout

- Logged-in cart + guest cart, persisted across sessions
- Voucher codes applied at cart with live recalculation
- Automated abandoned cart and checkout recovery emails
- Multiple payment methods in one flow:
 - Stripe Checkout (cards, Apple Pay, Google Pay)
 - Bank Transfer with proforma invoice (PDF with due date) emailed automatically
 - Cash on Delivery with admin manual confirmation

- Shipping cost calculation (per-zone, weight-based)
- Address management (multiple saved addresses, separate billing/delivery)
- T&C and GDPR consent enforcement (frontend + backend defense in depth)

Account & post-purchase

- Self-service registration with email confirmation
- Forgot-password flow with rate-limited reset links (24h expiry)
- Order history with full status tracking (PLACED -> CONFIRMED -> PACKING -> SHIPPED -> DELIVERED)
- Public invoice download (anyone-with-link, audit-logged, rate-limited, status-aware expiry)
- Self-service refund requests with item-level granularity (full or partial)
- Storno (credit note) auto-generated and emailed on refund approval
- Newsletter subscribe/unsubscribe (one-click via signed token in email)
- GDPR account anonymization (Article 17 compliant, audit-logged) — purges PII while preserving fiscal records to satisfy retention requirements (configurable retention window per jurisdiction)

Compliance & legal

- Cookie consent (functional cookies only by default)
- Configurable Privacy / Terms / Returns / Cookies pages — content lives in editable templates, translatable to any market
- Failed-payment recovery emails before configurable order auto-cancel
- Configurable return window (default 14 days, EU consumer-rights baseline)

ADMIN PANEL

Catalog management

- Product CRUD with multi-photo upload (cloud-hosted, CDN-served)
- Bulk CSV import for products / categories / producers with fuzzy entity matching (auto-corrects mistyped category/producer names; flags unmatched rows for admin review)
- CSV export for backup or off-platform analysis
- Category tree management with parent-child links, drag-friendly bulk operations
- Producer / brand management with linked product lists

- Per-product SEO fields (meta title, description, keywords) with live AI evaluation panel for search display, LLM readability, and AI crawler discoverability
- Discount-exempt category flag (for items that shouldn't participate in cart-level promotions)

Order operations

- Full order lifecycle dashboard with tab-based filtering by status
- Order details with line items, customer info, payment timeline
- Status transitions enforced by validator (no illegal state changes)
- Manual edits to order contents pre-shipment (add/remove/change quantity)
- Tracking URL attachment on SHIPPED status
- Email notifications fire automatically per status change (CONFIRMED, PACKING, SHIPPED, DELIVERED, refund states)

Refund & storno

- Per-item refund granularity (full or partial)
- Online payment refunds processed automatically — idempotent, webhook-driven completion
- Bank Transfer / COD refunds tracked manually with full audit trail
- Storno (credit note) PDF auto-generated on refund — separate fiscal document with line items, discount distribution per line, sign convention reconciles cleanly against the original invoice
- Shipping policy: shipping NOT refunded post-delivery (EU consumer-law standard for return-after-receipt); IS refunded for pre-shipment cancellations
- Public storno download link emailed to customer (same audit pattern as invoices)

Marketing & promotions

- Voucher codes (manual code input by customer at cart)
- Discount rules (auto-applied based on conditions: min order, category-specific, percentage or fixed amount)
- Per-user voucher limits (e.g., one-use welcome voucher with 5% off)
- Email broadcast campaigns plus cart, checkout, and failed-payment recovery flows, all by email
- Discount/sale notifications for wishlist items dropping in price (opt-in)

Storefront branding

- Logo, header banner, footer banner, page background — all upload/replace/delete from admin
- Currency selector (EUR/RON/USD/GBP)
- Configurable return-days policy

- Vacation mode (with custom message displayed to customers)
- Stock-management toggle (production deploy currently runs without stock — admin manually confirms each order to avoid oversell)

Customer support

- Product feedback moderation with configurable profanity filter
- Customer search with full-text + filter combinators
- Per-customer order history view
- Notification panel for system incidents (email failures, payment webhook anomalies, storage outages, etc.) — email alert + in-panel triage

Reporting

- Sales analytics by period
- Top products by volume / revenue
- Customer lifetime value
- Upload history audit trail (who imported what CSV when, with rejected/corrected subset CSVs preserved in cloud storage)
- Accounting CSV exports — analytics-ready data feed. One click exports order-level totals and per-product order lines for the selected period, including invoice number, order id, dates, prices, totals before/after discount, currency, payment method, buyer + seller details, IBAN, payment terms, and notes.

FISCAL & ACCOUNTING

- Invoice PDF generation with EU-style fiscal-compliance fields: company registration number, tax / VAT ID, registered address, IBAN, bank name, VAT payer status, VAT rate, sequential numbering, due dates for Bank Transfer
- Invoices generated automatically at the right moment per payment method:
 - Stripe: post-payment confirmation
 - Bank Transfer: at admin confirmation (proforma with due date) -> marked paid when admin confirms transfer received
 - COD: at SHIPPED status
- Storno (credit note) for every refund — separate document, separate numbering, linked to original invoice

- Configurable invoicing fields per shop (company name, tax ID, registration number, IBAN, default currency, invoice prefix, notes)
- Configurable VAT-payer / non-VAT-payer status and VAT rate per shop
- e-Factura-compatible XML generated alongside fiscal invoices for manual upload into the government portal
- Provider abstraction remains available for future direct integration with any e-invoicing provider

ARCHITECTURE & ENGINEERING

The platform is built on enterprise-grade, production-tested open-source technologies. Modular architecture separates admin, storefront, and core services while shipping as a single deployable unit. The database layer uses explicit SQL (no ORM magic, no hidden queries). Caching, session management, and rate limiting run on an in-memory data store. Sensitive tokens are encrypted at rest before reaching the database. All files (product images, invoices, CSV backups, branding assets) live in cloud object storage with CDN delivery.

Engineering decisions worth highlighting

Idempotency everywhere that matters.

Payment webhooks, refund triggers, CSV imports (re-running doesn't duplicate), category-link uploads. Re-running any operation is safe by default.

Audit trails on every sensitive flow.

Compliance-grade record-keeping built in, not bolted on. Every sensitive flow — uploads, invoice access, account anonymization, system incidents — is audit-logged.

Fuzzy entity matching on bulk imports.

When admin uploads a CSV of products with category/producer names, the system auto-matches against existing entities (exact match), suggests close matches via fuzzy distance scoring (high-confidence — admin verifies via downloadable subset CSV pre-populated with the corrected entity names), and flags unmatched rows (admin fixes & re-imports). No silent failures, no orphan data, no manual ID lookup. Same pattern wraps category-link uploads and producer imports.

Broadcast email engine, in-house.

Built-in SMTP integration with per-recipient delivery tracking, audience segmentation (all-active / newsletter-opted / has-orders), unsubscribe via signed token (one-click, no login required), campaign analytics, abandoned cart / checkout recovery, and failed-payment follow-ups. Most platforms hand this off to third-party email services and pay per-contact monthly — here it's built into the admin panel and costs only direct sending fees (~\$0.10 / 1k emails).

Incident notification system, built-in.

Server-side errors that matter (email delivery failures, payment webhook anomalies, storage outages, invoice generation failures) surface in an admin in-panel triage view and fire an alert email to the operator. No external monitoring subscription needed for the operator-grade visibility a single-tenant deploy actually uses.

Token encryption at rest.

Sensitive tokens (auth tokens, password reset tokens, public invoice access tokens) are encrypted before hitting the database. A database dump on its own is not enough to compromise tokens. Session tokens use asymmetric key signing, not shared-secret hashing.

Event-driven refund flow.

A single internal event triggers all refund side-effects (storno generation, email, status transitions) regardless of whether the refund came from a payment webhook, admin manual approval, or user self-service cancellation. One implementation, one test surface.

Defense in depth on security & UX.

Backend validates every order state transition (no illegal state changes); frontend hides ineligible actions; rate limiting on login + public links; CSRF on state-changing endpoints; user-facing error messages always sanitized in the storefront's locale, with technical details kept only in logs + incident notifications.

Migration-friendly design.

Built explicitly with the assumption that merchants are coming from PrestaShop / WooCommerce / Magento. CSV import schema designed to ingest realistic exports. Silent password rehash supported for migrating users from legacy platforms (users log in with their existing passwords on day one without seeing a forced reset).

WHAT YOU DON'T HAVE TO BUILD (PER CLIENT)

Things an agency typically pays developer-hours to bolt on per project — already in the box here:

CONCERN	BUILT IN
Email broadcast + segmentation + unsubscribe	Built-in engine, no third-party subscription per merchant
Cart / checkout / payment recovery	Automated email sequences for abandoned carts, abandoned checkout, and failed payments
Operational error visibility	Notification panel + admin triage view + email alerts
Bulk import without manual data cleanup	Fuzzy matcher for category/producer names
Idempotent payment + refund webhooks	Event-driven pipeline, safe under re-delivery
GDPR Article 17 (right to erasure)	One-click anonymization, audit-logged, fiscal records preserved
Public document download (invoice, storno)	Signed link + audit log + rate limit + status-aware expiry
EU-style fiscal compliance	Tax ID / company registration / IBAN / VAT settings / sequential numbering / due dates / storno credit notes
e-Factura manual upload	e-Factura-compatible XML generated for accountant or merchant portal upload
Token encryption at rest	Encrypted before database storage

CONCERN	BUILT IN
Session + rate limit + token blacklist	In-memory caching layer, drop-in
Object storage with CDN	Cloud storage wired (invoices / photos / CSVs / branding)
CSV export + re-import roundtrip	Schema-aligned, used in production import flows
Accounting CSV exports	Order-level totals plus per-product order lines — directly consumable by accountant, spreadsheet, BI, or AI-driven sales analysis
Auto-PDF for invoices + storno	A4-layout PDF generation built in

The point isn't "look how many features" — it's that each of these is a 1–4 week build per client if done from scratch. Pre-built means your billable scope can stay on the parts the client actually asked for (branding, content, integrations, custom logic) instead of the plumbing every shop needs.

WHAT IT DOES NOT DO (YET)

Honesty section. This isn't Shopify — different choices were made.

- Variants (size x color x capacity -> SKU per combo) — single-SKU model currently. Architectural priority for next sprint (5-7 days). Designed: master product + variant table with flexible specification attributes (universal across verticals: laptops/fashion/appliances).
- Multi-tenant — one shop per deploy currently. Tenant isolation refactor possible but not in scope.
- Marketplace (multi-seller, royalties, payouts) — out of scope; different product entirely.
- Subscriptions / recurring billing — recurring payment API not wired.
- Multi-language storefront UI — internationalization framework is configured and content lives in editable templates, but the runtime locale switcher is not wired yet (one default language ships per deploy). Adding more locales is a content task.
- Direct native government e-invoicing portal submission — jurisdiction-specific. For now the platform generates e-Factura-compatible XML for manual upload; provider abstraction remains in place to plug in any country's portal.

WHERE IT FITS COMMERCIALLY

The platform was built to be deployed as a custom code project for a new merchant — not as a SaaS product. An agency or partner takes the codebase, brands and configures it for the client, and delivers a self-hosted ecommerce backend the client owns outright. Per-deploy economics, not per-seat.

Primary use case — agency-delivered custom build for a new merchant

- Greenfield ecommerce launch where the client wants their own code, not a SaaS rental
- Agency keeps configuration + branding + content + project-management revenue at its own rates — that's the agency's main margin
- Client gets a production-grade backend (catalog, payments, refunds, fiscal docs, GDPR, audit) without paying SaaS rent forever
- EU-fiscal-ready out of the box; localization to any market is a content task, not engineering

Secondary use case — migration off SaaS / aging stacks

- Shopify merchants tired of per-transaction fees and customization walls
- PrestaShop 1.6/1.7, Magento 1, custom PHP / ASP.NET shops where the platform has hit its ceiling
- CSV import schema designed to ingest realistic exports; silent password rehash for legacy platform migrations means users keep their old passwords on day one

Why it's a good base for an agency

- One codebase, one ops surface — repeated deploys don't compound your maintenance load
- Modular architecture means you can carve out a service when a client genuinely needs it, without architecting for it on day one
- Fiscal compliance + GDPR + audit trails are already built — you don't pay developer-hours to rebuild them per client
- Provider abstractions mean swapping payment processors, or plugging in a country-specific accounting / e-invoicing provider, is a configuration change, not a rewrite

Not the right tool for

- Headless-only setups (this comes with a storefront)
- Marketplace operators (multi-seller is out of scope)
- Brand-new solo merchants under €50k GMV who want zero-touch hosting — Shopify Basic is cheaper to start at that scale

OPERATING COST (ROUGH ORDER OF MAGNITUDE)

For a single-shop deploy:

LINE ITEM	MONTHLY
Hosting (VPS, can be a single 4GB/2vCPU box for SMB volume)	€15–30
Cloud storage + CDN	€1–10
Email sending (transactional + monthly campaign of 10k emails)	€1–5
Stripe fees	per-transaction only (1.4% + €0.25 EU cards typical)
Total infra	~€20–50/mo + Stripe

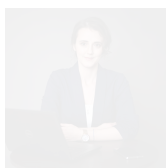
Compare to Shopify Basic (\$39/mo) + Shopify transaction fees (2.0% if not using Shopify Payments) + per-app subscriptions. For a merchant doing €50k+/mo GMV, the difference compounds quickly.

WHERE I AM IN THE BUILD

- Storefront + admin core (~95% done, polishing in progress)
- Stripe + BT + COD payment flows end-to-end
- Refund + storno (EU fiscal-grade credit notes)
- Email engine with broadcast campaigns
- GDPR compliance (anonymization, public invoice expiry, audit logs)
- First-merchant launch in 2-3 weeks (~40 orders/mo baseline)
- Post-launch: variants + per-product specs (5-7 day sprint), then second client onboarding

PRICING NOTES & SCOPE

Each engagement is structured as initial license + recurring maintenance retainer + custom dev hours. Operational costs (hosting, email, payment processing) are always merchant-paid passthrough. Specifics shaped by scope.



WHAT I'M LOOKING FOR

IOANA DINU · E-COMMERCE SOLUTIONS ARCHITECT

An agency / studio / freelance partner who delivers ecommerce builds as custom code projects to their clients (not as SaaS resale) and wants a proven backend to build on instead of starting from zero each time.

The ideal partner is already pitching custom-API / custom-code work to new clients — typically merchants who specifically want to own their stack rather than rent it. EU market focus (fiscal-compliance fields are built in; localization to any language is a content task).

Open to:

- Agency-partner referral — you bring the client; client contracts directly with me at standard rates; you receive recurring commission for the life of the contract; you keep all delivery-scope revenue (branding, configuration, content, project management) at your own rates
- Direct sale + onboarding for a merchant who finds me directly
- Migration projects off Shopify / Presta / Magento as a specific service line on top of either path
- Open to discussing other arrangements that fit your business model

Happy to demo end-to-end on a video call — full flow takes ~20 minutes.

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